

EXP NO 1:

```
[1]
✓ 3s from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

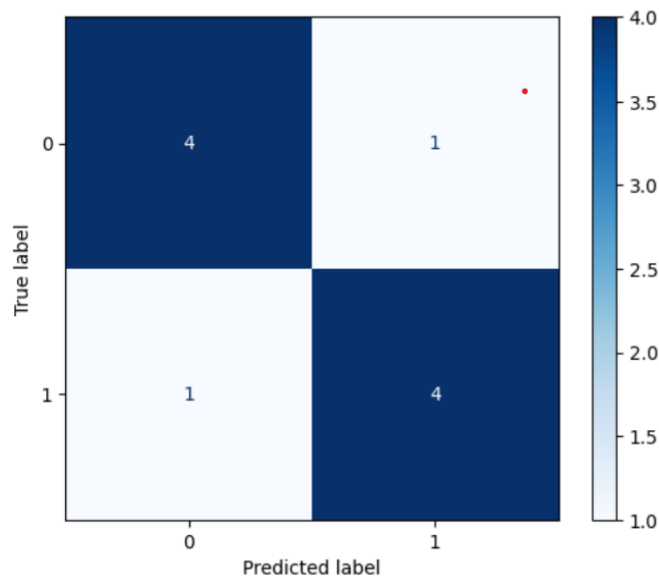
y_true = [1,0,1,1,0,1,0,0,1,0]
y_pred = [1,0,1,0,0,1,1,0,1,0]

cm = confusion_matrix(y_true, y_pred)

print("Confusion Matrix:")
print(cm)

disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap="Blues")
plt.show()
```

▼ ... Confusion Matrix:
[[4 1]
 [1 4]]



EXP NO 2

```
[1]
✓ 3s import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, f1_score, precision_score, recall_score, classification_report, confusion_matrix

wine = load_wine()
data = pd.DataFrame(data=wine.data, columns=wine.feature_names)
data['Target'] = wine.target
data = data[data['Target'] != 2]
x = data.drop('Target', axis=1)

y = data['Target']
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3, random_state=1)
model = DecisionTreeClassifier(random_state=1)
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
accuracy = accuracy_score(y_test, y_pred)
print("accuracy:", accuracy)
class_report = classification_report(y_test, y_pred, target_names=wine.target_names[:2])
print("Classification Report:\n", class_report)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1:.2f}")
conf_matrix = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8,6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='PuBuGn', xticklabels=wine.target_names[:2], yticklabels=wine.target_names[:2])
plt.xlabel("predicted label")
plt.ylabel("true label")
plt.title("confusion matrix")
plt.show()
```

```

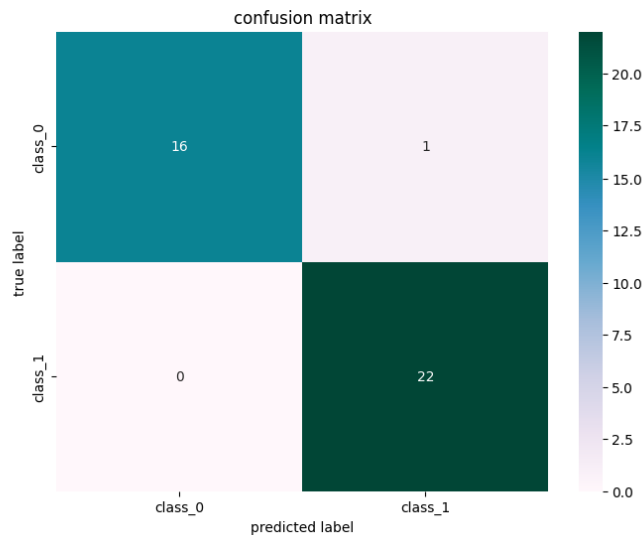
*** accuracy: 0.9743589743589743
Classification Report:
              precision    recall  f1-score   support

   class_0       1.00        0.94        0.97         17
   class_1       0.96        1.00        0.98         22

 accuracy
macro avg       0.98        0.97        0.97         39
weighted avg    0.98        0.97        0.97         39

Precision: 0.96
Recall: 1.00
F1 Score: 0.98

```

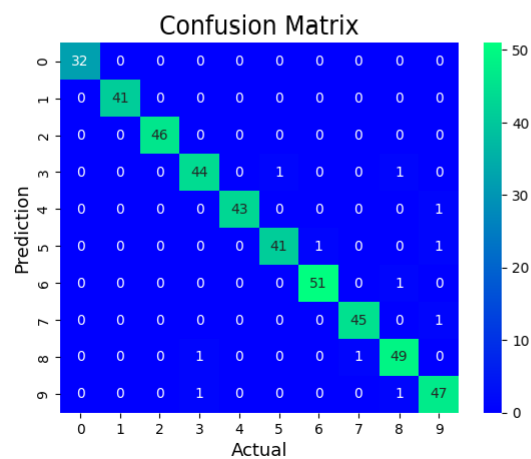


EXP NO 3

```

[6] 2s
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
X, y = load_digits(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25)
clf = RandomForestClassifier(random_state=23)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm,
            annot=True,
            fmt='g', cmap="winter")
plt.ylabel('Prediction', fontsize=13)
plt.xlabel('Actual', fontsize=13)
plt.title('Confusion Matrix', fontsize=17)
plt.show()
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy :", accuracy)

```



Accuracy : 0.9755555555555555

EXP NO 4



EXP NO 5

```

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

iris = load_iris()
data = pd.DataFrame(data=iris.data, columns=iris.feature_names)
data['species'] = iris.target
print(data.head())
X = data[['sepal length (cm)']]
y = data[['petal width (cm)']]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
print(f'Mean Squared Error on test set: {mse:.2f}')
plt.figure(figsize=(10, 6))
plt.scatter(X_test, y_test, color='r', marker='*', label='Actual')
plt.plot(X_test, y_pred, color='y', linewidth=3, label='Predicted')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Petal Width (cm)')
plt.title('Linear Regression: Sepal Width vs Sepal Length')
plt.legend()
plt.show()

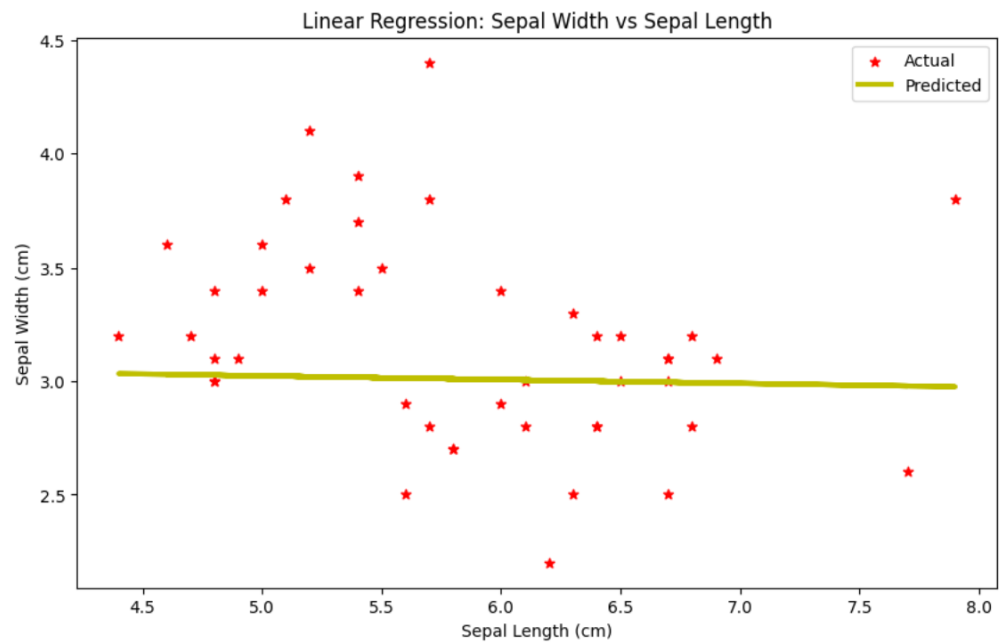
new_sample = pd.DataFrame([[5]], columns=['sepal length (cm)'])
predicted_width = model.predict(new_sample)
print(f'The predicted sepal width for sepal length {new_sample.values.tolist()} is {predicted_width[0]:.2f} cm')

```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
...					
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

	species
0	0
1	0
2	0
3	0
4	0

Mean Squared Error on test set: 0.23



The predicted sepal width for sepal length `[[5]]` is 3.02 cm